Lab 2: Protocol Layers and Security Design

Network System Design | SE026 | 2026-27

Objective

Use packet evidence to explain protocol layers and propose access boundaries for a building network.

1. Software and files

Install a stable release of Wireshark for your operating system from https://www.wireshark.org/download.html. Have Week2-Teaching-Trace.pcap available on your computer.

This lab reads a saved capture file. Live capture is not required. Opening the file does not need an Internet connection.

Open the trace

1. Start Wireshark. Find the version under Help > About Wireshark, or the application About menu on macOS.

2. Choose File > Open and select Week2-Teaching-Trace.pcap.

3. Locate Packet List, Packet Details and Packet Bytes.

If the file does not open, download it again and check that you selected the .pcap file rather than the ZIP archive.

About the trace

The example trace contains DNS and plain HTTP/1.1 at a client-side Ethernet observation point. Its documentation addresses and .test name are illustrative. It does not measure campus traffic.

Record	Your observation
OS / Wireshark version	_____
Capture filename	_____
Unfiltered packet count	_____

2. DNS and protocol fields

You may consult the lecture material on DNS and protocol layers. Enter dns in the display-filter bar and press Enter. Inspect a query and its matching response. Clear the filter before examining other traffic.

Observation	Your evidence and explanation
DNS query / response frame numbers	_____
Query name and record type	_____
Query source / destination IPv4	_____
UDP source / destination port	_____
Returned address	_____

Explain the name-lookup protocol, the protocol carrying this exchange and the layer supplying IPv4 delivery. Does this trace prove that every DNS exchange uses UDP? Explain the evidence limit.

3. HTTP request and encapsulation

Apply the http display filter. Select the request and expand Ethernet II, IPv4, TCP and HTTP in Packet Details. The lecture material on encapsulation may help you interpret the fields.

Layer evidence	Your field and meaning
Ethernet source / destination MAC	_____
IPv4 source / destination address	_____
TCP source / destination port	_____
HTTP method / Host / resource	_____
Response frame / status	_____

Right-click the request and choose Follow > TCP Stream. Read the exchange, then close the view and clear the resulting stream filter. Explain which text is visible and which layers carry it. Use frame numbers to identify your examples.

4. Forwarding and trust boundaries

You may consult the lecture material on forwarding, trust boundaries and security controls.

Design scenario

A student lab and a finance office share a building network. Both require Internet access. Only finance users require access to the payroll service. Treat this as a fictional design exercise.

1. Draw the endpoints, access devices, gateway and service location. Label student, finance and management trust groups.

2. Mark the permitted service paths and one path to restrict.

3. Propose a control and explain how you would verify it.

4. Identify a confidentiality threat illustrated by plain HTTP and propose a suitable control.

Requirement / threat	Proposed control	Verification / limit
_____	_____	_____
_____	_____	_____
_____	_____	_____

Forwarding explanation

Explain what ordinarily changes in the Ethernet frame at a router and what remains in the IP source and destination addresses when no NAT, tunnel or proxy intervenes. The trace shows one client-side link; label the next-link frame as a conceptual prediction.

A clear hand-drawn or digital diagram is sufficient. This activity asks for a design and a verification plan; device configuration is not required.

Notes and diagram labels

_____

_____

_____

_____

5. Reflection

1. Explain how an HTTP message fits inside TCP, IP and Ethernet.

_____

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_____

2. Why can the destination MAC identify a different device from the remote server named by the IP destination?

_____

_____

_____

3. Which security goal does your proposed control address?

_____

_____

_____

4. What evidence is still needed before implementing the design?

_____

_____

_____

Further help

Wireshark User Guide: https://www.wireshark.org/docs/wsug_html_chunked/ChapterWork.html

Display filters: https://www.wireshark.org/docs/wsug_html_chunked/ChWorkBuildDisplayFilterSection.html